(KTXFI1EBNF) Physics I. Practice

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May 4, 2026

Current Updates and Announcements

- May 4, and go...
- Homework No. 3 is now available tonight. The deadline is May 10, 2026 at midnight.
- I promised a sample exam. I provided $\sim$100 questions on Moodle.
- Today there is only practice of three different type of exercises.

Exercises I

- **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
- **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.
- **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
- **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
- **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).

Exercises II

- **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.
- **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
- **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

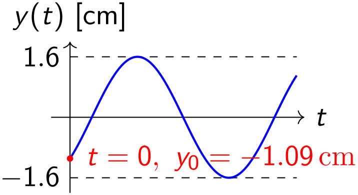
Oscillations and Waves I

1. Given the displacement-time function of the oscillation:

$$y(t) = 1.6 \sin(1.3t - 0.75)\, cm.$$

At $t = 0\, s$, determine:

a) displacement,
b) velocity,
c) acceleration.



Convert amplitude: $1.6\ \mathrm{cm} = 0.016\ \mathrm{m}$

a) $y(0) = 0.016\,\mathrm{m} \sin(-0.75) = -0.0109\ \mathrm{m}$

b) Velocity: $v(t) = \frac{dy(t)}{dt} = 0.016\,\mathrm{m} \cdot 1.3\,\mathrm{s}^{-1} \cos(1.3t - 0.75)$
$v(0) = 0.0208\,\mathrm{m/s} \cos(-0.75) = 0.0152\ \mathrm{m/s}$

c) Acceleration:
$a(t) = \frac{dv(t)}{dt} = -0.016\,\mathrm{m} \cdot (1.3\mathrm{s}^{-1})^2 \sin(1.3\,\mathrm{s}^{-1}\ t - 0.75)$
$a(0) = 0.0184\ \mathrm{m/s}^2$

2. Given the sum of two one-directional oscillation as the following $x(t) = 3\sin(\omega t) + 4\cos(\omega t)$. This describes the displacement - time function of the oscillation, where t represents time, ω denotes the circular frequency of the oscillation, and ω is a constant.
   a) Prove that the above x(t) oscillation, given by the displacement-time function, performs simple harmonic motion.
   b) What is the value of the resultant oscillation given by the above displacement-time function?

   a) First, let's take the first and second time derivatives of $x(t)$!

$$
\begin{aligned}
\frac{dx(t)}{dt} &= \dot{x}(t) = 3\omega\cos(\omega t) - 4\omega\sin(\omega t) \\
\frac{d^2x(t)}{dt^2} &= \ddot{x}(t) = -3\omega^2\sin(\omega t) - 4\omega^2\cos(\omega t)
\end{aligned}
$$

Oscillations and Waves III

After that, let's form the ratio of $\frac{\ddot{x}(t)}{x(t)}$

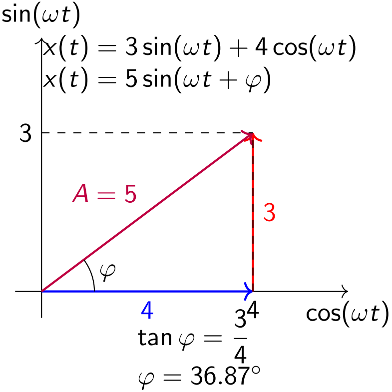
$$\begin{aligned}
\frac{\ddot{x}(t)}{x(t)} &= \frac{-3\omega^2\sin(\omega t) - 4\omega^2\cos(\omega t)}{3\cos(\omega t) - 4\sin(\omega t)} = \frac{\omega^2\left[-3\sin(\omega t) - 4\cos(\omega t)\right]}{3\cos(\omega t) - 4\sin(\omega t)} = -\omega^2 \\
\frac{\ddot{x}(t)}{x(t)} &= -\omega^2 \\
\ddot{x}(t) &= -\omega^2 x(t)
\end{aligned}$$

where $-\omega^2$ is constant. This is precisely the equation of the harmonic motion. Thus, the oscillation is harmonic.

b) To calculate the amplitude, we use the amplitude formula:

$$A = \sqrt{A_1^2 + A_2^2 + 2 \cdot A_1 \cdot A_2 \cdot \cos\left(\varphi_1 - \varphi_2\right)}$$

Oscillations and Waves IV



Calculating the phase difference:

$x(t) = 3\cdot\sin(\omega t) + 4\cdot\cos(\omega t) = 3\cdot\sin(\omega t) + 4\cdot\sin\left(\omega t + \frac{\pi}{2}\right)$

$\cos(\alpha) = \sin\left(\alpha + \frac{\pi}{2}\right)$

The phase difference: $\varphi_1 = \omega t$, $\varphi_2 = \omega t + \frac{\pi}{2}$.

$$\varphi_1 - \varphi_2 = -\frac{\pi}{2}$$

$A = \sqrt{A_1^2 + A_2^2 + 2\cdot A_1\cdot A_2\cdot\cos(\varphi_1 - \varphi_2)} =$
$\sqrt{3^2 + 4^2 + 2\cdot 3\cdot 4\cos\left(-\frac{\pi}{2}\right)}$

Therefore, the amplitude is A = 5.

Oscillations and Waves V

3. Consider three one-directional oscillations with equal frequencies, whose amplitudes and initial phases are given as follows:

$$A_1 = 7 \cdot 10^{-3}, \mathrm{m} \quad , \quad \varphi_1 = \pi/3$$
$$A_2 = 5 \cdot 10^{-3}\, \mathrm{m} \quad , \quad \varphi_2 = 5\pi/6$$
$$A_3 = 2 \cdot 10^{-3}\, \mathrm{m} \quad , \quad \varphi_3 = 23\pi/18$$

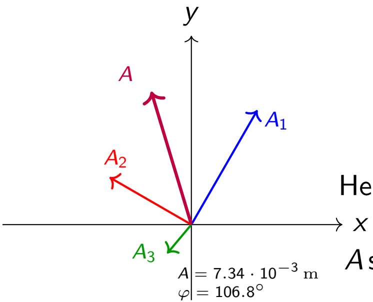
Let's combine these three oscillations. Calculate the amplitude and initial phase of the resultant vibration!

The amplitude of the resultant oscillation - sum of oscillations - can be calculated as follows:

$$x(t) = A \sin(\omega t + \varphi)$$

Oscillations and Waves VI

Using vector addition:



$$A \sin\varphi = \sum_i A_i \sin\varphi_i$$

$$A \cos\varphi = \sum_i A_i \cos\varphi_i$$

Hence,

$$A \sin\varphi = 7 \cdot 10^{-3} \sin\frac{\pi}{3} + 5 \cdot 10^{-3} \sin\frac{5\pi}{6} + 2 \cdot 10^{-3} \sin\frac{23\pi}{18}$$

$$A \cos\varphi = 7 \cdot 10^{-3} \cos\frac{\pi}{3} + 5 \cdot 10^{-3} \cos\frac{5\pi}{6} + 2 \cdot 10^{-3} \cos\frac{23\pi}{18}$$

Oscillations and Waves VII

Numerically:

$$\begin{aligned} A\sin\varphi &= 0.007 \\ A\cos\varphi &= -0.002 \end{aligned}$$

Dividing:

$$\tan\varphi = \frac{A\sin\varphi}{A\cos\varphi} = \frac{0.007}{-0.002} = -3.5$$

Thus:

$$\varphi \approx 105.945^\circ$$

or

$$\varphi \approx 1.85 \ \mathrm{rad}$$

Oscillations and Waves VIII

Using:

$$A \sin \varphi = 0.007$$

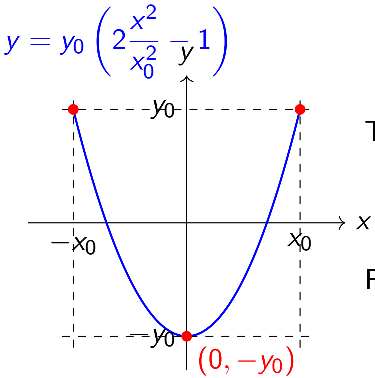
we obtain:

$$A \approx 7.3 \cdot 10^{-3} \mathrm{~m}$$

Oscillations and Waves IX

4. Calculate and determine the curve described by the resultant of the perpendicular vibrations given by the equations:

$$
\begin{aligned}
x(t) &= x_0 \cos(\omega t), \\
y(t) &= y_0 \cos(2\omega t).
\end{aligned}
$$



The perpendicular oscillations are given by

$$
\begin{aligned}
x(t) &= x_0 \cos(\omega t) \\
y(t) &= y_0 \cos(2\omega t).
\end{aligned}
$$

First, express $\cos(\omega t)$ from the first equation:

$$\cos(\omega t) = \frac{x}{x_0}.$$

Oscillations and Waves X

Next, use the trigonometric identity

$$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = \cos^2\alpha + \left(\cos^2\alpha + \sin^2\alpha\right) - \sin^2\alpha - 1 = 2\cos^2\alpha - 1,$$

where $\sin^2\alpha + \cos^2\alpha = 1$

Applying this identity to the second equation:

$$y = y_0\left(2\cos^2(\omega t) - 1\right).$$

Substitute

$$\cos(\omega t) = \frac{x}{x_0}$$

into the equation:

$$y = y_0\left(2\left(\frac{x}{x_0}\right)^2 - 1\right).$$

Oscillations and Waves XI

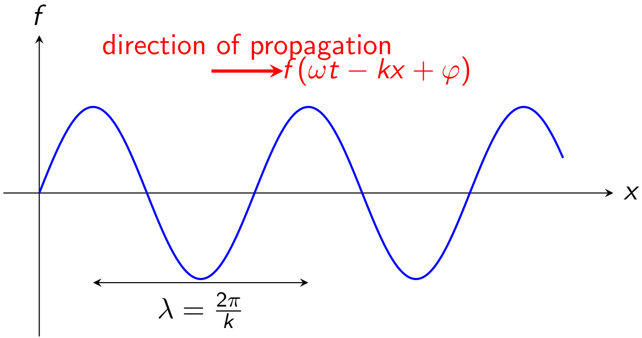
Therefore, the equation of the resultant curve is

$$\boxed{y = y_0 \left(2\frac{x^2}{x_0^2} - 1 \right)}$$

which is the equation of a parabola.

Oscillations and Waves XII

5. Let $f(\omega t \pm kx + \varphi)$ be a function whose argument is at least twice differentiable, continuous, and arbitrary, where ω: constant circular frequency, k : wavenumber (constant), φ initial phase (constant), t : time, x spatial coordinate. Prove that the function f described above is a solution to the one-dimensional wave equation.



Oscillations and Waves XIII

The one-dimensional wave equation is

$$\frac{\partial^2 \psi}{\partial t^2} = c^2 \frac{\partial^2 \psi}{\partial x^2},$$

where c is a constant, the wave velocity.
Let $\psi = f(\omega t \pm kx + \phi)$. Take the derivatives. Time derivatives:

$$\begin{aligned} \frac{\partial f}{\partial t} &= \omega f'(\omega t \pm kx + \phi) \\ \frac{\partial^2 f}{\partial t^2} &= \omega^2 f''(\omega t \pm kx + \phi) \end{aligned}$$

Oscillations and Waves XIV

Spatial derivatives:

$$\frac{\partial f}{\partial x} = \pm k f'(\omega t \pm kx + \phi)$$
$$\frac{\partial^2 f}{\partial x^2} = k^2 f''(\omega t \pm kx + \phi)$$

Substituting into the wave equation:

$$\omega^2 f''(\omega t \pm kx + \phi) = c^2 k^2 f''(\omega t \pm kx + \phi)$$

Using the $\omega = kc$, $\omega^2 = k^2 c^2$ dispersion relations,

$$\omega^2 f''(\omega t \pm kx + \phi) = \omega^2 f''(\omega t \pm kx + \phi)$$

Hence both sides are identical, proving the statement.
Wave speed: $v = \frac{\omega}{k}$ $[\mathrm{m/s}]$

Oscillations and Waves XV

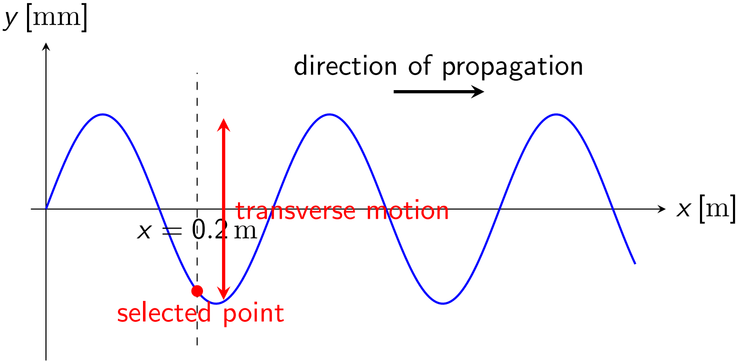
6. The displacement of a wave as a function of time is given by

$$y(x,t) = 0.27\sin(12x - 500t)\,mm,$$

where x is in meters and t is given in seconds. Determine at the point $x = 0.2$ meters in the case of a transverse wave:

a) its velocity,
b) its acceleration!
c) What is the displacement of the point at $t = 4s$ seconds?

Oscillations and Waves XVI



a) Velocity is $v = \frac{\partial y}{\partial t}$, $v = -0.27 \cdot 500 \cos(12\,\mathrm{m}^{-1}x - 500\,\mathrm{s}^{-1}t)\ \frac{\mathrm{mm}}{\mathrm{s}}$

At $x = 0.2\,\mathrm{m}$: $$v = -0.135 \cos(2.4 - 500\,\mathrm{s}^{-1}t)\ \frac{\mathrm{m}}{\mathrm{s}}$$

Oscillations and Waves XVII

b) Acceleration is $a = \frac{\partial v}{\partial t}$, therefore,

$$a = -0.135 \cdot 500 \cdot \sin(2.4 - 500\, s^{-1} t)\mathrm{m/s^2} = -67.5 \sin(2.4 - 500\,\mathrm{s}^{-1} t)\ \frac{\mathrm{m}}{\mathrm{s}^2}$$

c) Displacement of the point at t = 4s:

$$y = 0.27 \cdot 10^{-3}\mathrm{m} \sin(12\,\mathrm{m}^{-1} \cdot 0.2\,\mathrm{m} - 500\,\mathrm{s}^{-1} \cdot 4\,\mathrm{s}) \approx 0.118\ \mathrm{mm}$$

Convert amplitude: $0.27\ \mathrm{mm} = 2.7 \cdot 10^{-4}\ \mathrm{m}$

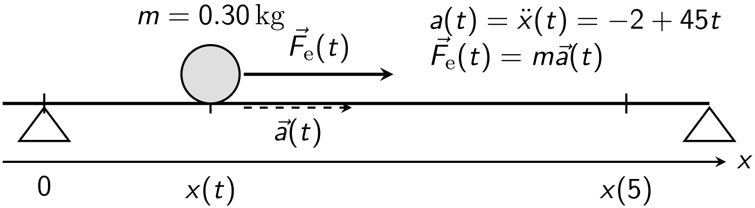
Rigid Bodies I

1. According to measurements, the position (in meters) of a body with a mass of 300 g along the x-axis is given by the following function:

$$x(t) = -t^2 + 7.5t^3,$$

where t represents the time in seconds. Determine the resultant force acting on the body as a function of time! What is the magnitude of the resultant force at t=5 s?

$$x(t) = -t^2 + 7.5t^3$$
$$a(t) = \ddot{x}(t) = -2 + 45t$$
$$\vec{F}_\mathrm{e}(t) = m\vec{a}(t)$$



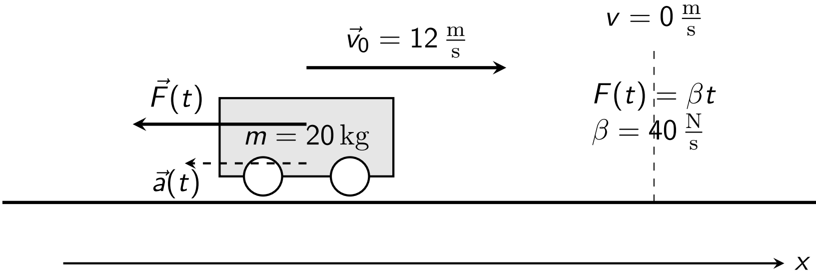
Rigid Bodies II

We know that $v(t) = \frac{dx(t)}{dt}$, $a(t) = \frac{dv(t)}{dt} = \frac{d^2x(t)}{dt^2}$, and $F(t) = m \cdot a(t)$.
Furthermore, $m = 300\,\mathrm{g} = 0.300\,\mathrm{kg}$, $x(t) = -t^2 + 7.5t^3\,\mathrm{m}$.

$$
\begin{aligned}
v(t) = \frac{dx(t)}{dt} = \frac{d}{dt}\left(-t^2 + 7.5t^3\right) &= -2t + 22.5t^2\,\frac{\mathrm{m}}{\mathrm{s}}, \\
a(t) = \frac{dv(t)}{dt} = \frac{d^2x(t)}{dt^2} &= -2 + 45t\,\frac{\mathrm{m}}{\mathrm{s}^2}, \\
F_\mathrm{e}(t) = m\,a(t) &= 0.300\,\mathrm{kg} \cdot (-2 + 45t)\,\frac{\mathrm{m}}{\mathrm{s}^2} = -0.6 + 13.5t\,\mathrm{N}, \\
F_\mathrm{e}(5\,\mathrm{s}) &= -0.6 + 13.5 \cdot 5\,\mathrm{N}, \\
F_\mathrm{e}(5\,\mathrm{s}) &= 66.9\,\mathrm{N}.
\end{aligned}
$$

Rigid Bodies III

2. A mass of 20 kg moves with a constant velocity at $12\,\mathrm{m/s}$. At some moment, we start to apply a force opposite to the direction of motion, which is not constant but increases steadily from zero at a rate of $40\,\mathrm{N/s}$. How much time elapses from the beginning of the force's effect until the body comes to a complete stop (decelerates)? A 20 kg mass moves with velocity 12 $\mathrm{m/s}$.



Rigid Bodies IV

The force can be expressed in the following form:

$$F = \beta t,$$

where $\beta = 40\,\mathrm{N/s}$.
According to Newton's second law:

$$F = ma \Rightarrow a = \frac{F}{m}.$$

Substituting the expression for the force:

$$a = \frac{F}{m} = -\frac{\beta t}{m}.$$

Rigid Bodies V

Therefore,

$$a = -\frac{\beta}{m} t,$$

and since

$$a = \frac{dv}{dt},$$

we obtain

$$\frac{dv}{dt} = -\frac{\beta}{m} t.$$

Rigid Bodies VI

Integrating both sides:

$$\int \frac{dv}{dt}\, dt = \int -\frac{\beta}{m} t\, dt$$

which gives

$$v = -\frac{\beta}{m} \int t\, dt = -\frac{\beta}{m} \cdot \frac{t^2}{2} + C.$$

Hence,

$$v = -\frac{\beta}{2m} t^2 + C.$$

Rigid Bodies VII

The integration constant is the initial velocity:

$$C = v_0,$$

where $v_0 = 12\,\mathrm{m/s}$ at $t = 0\,\mathrm{s}$ (beginning of the force action). Therefore,

$$v = v_0 - \frac{\beta}{2m}t^2.$$

When the object comes to a stop, $v = 0\,\mathrm{m/s}$.
Let the stopping time be denoted by τ. Then:

$$0\,\mathrm{m/s} = v_0 - \frac{\beta}{2m}\tau^2$$

Rigid Bodies VIII

hence,

$$\tau = \sqrt{\frac{2mv_0}{\beta}}.$$

Substituting the numerical values:

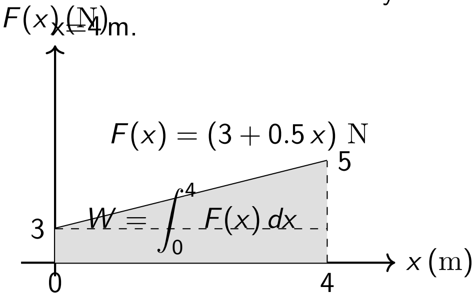
$$\tau = \sqrt{\frac{2mv_0}{\beta}} = \sqrt{\frac{2 \cdot 20\,\mathrm{kg} \cdot 12\,\mathrm{m/s}}{40\,\mathrm{N/s}}} = 3.5\,\mathrm{s}.$$

Rigid Bodies IX

3. A point-like object is subjected to a force in the x direction:

$$F(x) = (3 + 0.5\,x)\,\mathrm{N}.$$

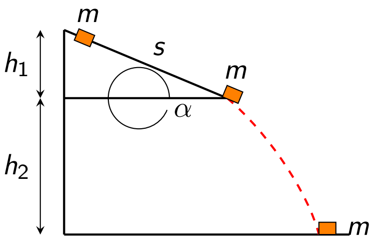
Determine the work done by the force as the mass point moves from x=0 m to x=4 m.



$$W = \int_g \vec{F} d\vec{s} = \int_0^4 (3 + 0.5\,x)\,dx\,\mathrm{J} = \left[3\,x + 0.25\,x^2\right]_0^4\ \mathrm{J} = 12\,\mathrm{J} + 4\,\mathrm{J} = 16\,\mathrm{J}$$

Rigid Bodies X

4. A 0.2 kg mass stone slides without initial velocity from the top edge of a frictionless roof as depicted in the figure. $\alpha = 40^\circ$, g=10 m$/s^2$, h1=3 m, h2=15 m
   a) What will be the magnitude of the velocity at the moment of impact with the ground?
   b) What will be the velocity upon impact with the ground if there is friction on the slope and the coefficient of kinetic friction is 0.4.



Rigid Bodies XI

4.1 A stone with mass $m = 0.2\,\mathrm{kg}$ starts to slide from rest from the top edge of a roof, as shown in the figure. The roof is inclined at an angle $\alpha = 40°$.
The following data are given:

$$g = 10\,\mathrm{m/s^2}, \qquad h_1 = 3\,\mathrm{m}, \qquad h_2 = 15\,\mathrm{m}.$$

(a) Assuming that the roof is frictionless, determine the magnitude of the velocity of the stone at the moment it reaches the ground.

(b) Determine the impact velocity if friction acts along the inclined roof and the coefficient of kinetic friction is $\mu = 0.4$.
Neglect air resistance.

The total vertical height difference between the initial and final positions is

$$H = h_1 + h_2 = 3\,\mathrm{m} + 15\,\mathrm{m} = 18\,\mathrm{m}.$$

Rigid Bodies XII

(a) **Frictionless motion:** Since there is no friction, mechanical energy is conserved. Initially, the stone is at rest, therefore its initial kinetic energy is zero. The conservation of mechanical energy gives

$$mgH + \frac{1}{2}mv_i^2 = mgh_f + \frac{1}{2}mv_f^2.$$

Because $v_i = 0\,\mathrm{m/s}$ and the ground level is chosen as $h_f = 0\,\mathrm{m}$, the equation becomes

$$mgH = \frac{1}{2}mv_f^2.$$

After cancelling the mass:

$$gH = \frac{1}{2}v_f^2.$$

Rigid Bodies XIII

Solving for the final velocity:

$$v_f = \sqrt{2gH}.$$

Substituting the numerical values:

$$v_f = \sqrt{2gH} = \sqrt{2 \cdot 10\,\mathrm{m/s^2} \cdot 18\,\mathrm{m}} = 18.97\,\mathrm{m/s}.$$

(b) **Motion with friction:** In this case, friction performs negative work on the stone. The conservation of mechanical energy becomes

$$mgH = W_{\mathrm{fric}} + \frac{1}{2}mv_f^2.$$

The friction force is

$$F_{\mathrm{fric}} = \mu N = \mu mg \cos\alpha.$$

Rigid Bodies XIV

The length of the inclined roof is

$$s = \frac{h_1}{\sin\alpha} = \frac{3\,\mathrm{m}}{\sin 40^\circ}.$$

The work done by friction is

$$W_{\mathrm{fric}} = F_{\mathrm{fric}} s = \mu m g \cos\alpha \cdot s.$$

Substituting this into the energy equation:

$$mgH = \mu m g \cos\alpha \cdot s + \frac{1}{2} m v_f^2.$$

After dividing by the mass and rearranging:

$$v_f = \sqrt{2gH - 2\mu g \cos\alpha \cdot s}.$$

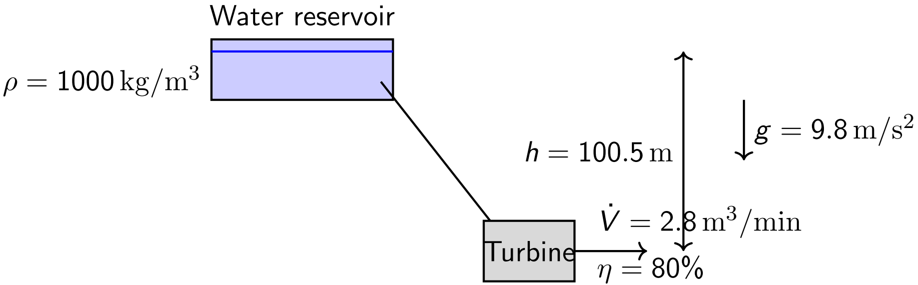
Rigid Bodies XV

Substituting the numerical values:

$$v_f = \sqrt{2 \cdot 10\,\mathrm{m/s^2} \cdot 18\,\mathrm{m} - 2 \cdot 0.4 \cdot 10\,\mathrm{m/s^2} \cdot \cos 40^\circ \cdot \frac{3\,\mathrm{m}}{\sin 40^\circ}} = 18.20\,\mathrm{m/s}.$$

Rigid Bodies XVI

5. From a water reservoir, water flows to a turbine located 100.5 m below. 2.8 m^3 of water passes through the turbine per minute. The efficiency of the turbine is 80%. Neglecting air resistance and the minimal kinetic energy of the water leaving the turbine, calculate the power output of the turbine! The density of water is 1000 kg/m^3, g=9.8 m/s^2.



Rigid Bodies XVII

The water flowing from the reservoir loses gravitational potential energy. This potential energy is converted into mechanical energy at the turbine. Since the turbine is not ideal, only a fraction of this energy appears as useful output energy. The mass of the water passing through the turbine in one minute is calculated from the density and the volume:

$$m = \rho V .$$

Substituting the numerical values gives:

$$m = \rho V = 1000\,\mathrm{kg/m^3} \cdot 2.8\,\mathrm{m^3} = 2800\,\mathrm{kg}.$$

The gravitational potential energy of this amount of water is:

$$E_{\mathrm{pot}} = mgh.$$

Rigid Bodies XVIII

Substituting the numerical values:

$$E_{\mathrm{pot}} = mgh = 2800\,\mathrm{kg} \cdot 9.8\,\mathrm{m/s^2} \cdot 100.5\,\mathrm{m}.$$

Therefore,

$$E_{\mathrm{pot}} = 2757720\,\mathrm{J}.$$

The input power of the turbine is the potential energy delivered to the turbine per unit time:

$$P_{\mathrm{in}} = \frac{E_{\mathrm{pot}}}{t}.$$

Therefore,

$$P_{\mathrm{in}} = \frac{2757720\,\mathrm{J}}{60\,\mathrm{s}}.$$

Rigid Bodies XIX

Hence the input power is:

$$P_{\mathrm{in}} = 45962\,\mathrm{W} \approx 46.0\,\mathrm{kW}.$$

The useful output power is obtained by multiplying the input power by the efficiency of the turbine:

$$P_{\mathrm{out}} = \eta P_{\mathrm{in}}.$$

Substituting the efficiency:

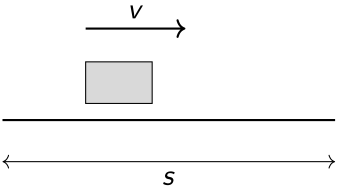
$$P_{\mathrm{out}} = \eta P_{\mathrm{in}} = 0.80 \cdot 45962\,\mathrm{W} = 36770\,\mathrm{W} \approx 36.8\,\mathrm{kW}.$$

Therefore, the useful power output of the turbine is approximately

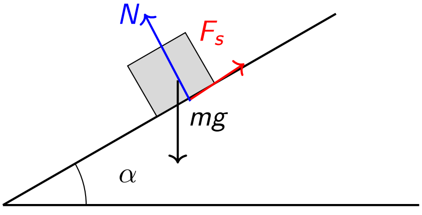
$$\boxed{P_{\mathrm{out}} = 36.8\,\mathrm{kW}.}$$

Rigid Bodies XX

6. A vehicle traveling with velocity v on a horizontal road decelerates uniformly over a distance s. What slope angle α will keep the vehicle at rest in its braked state on a similarly graded road? Neglecting air resistance.



Horizontal road



Inclined road

Rigid Bodies XXI

First, we determine the coefficient of static friction μ from the braking motion on the horizontal road. During braking, the vehicle moves with uniform deceleration. Its initial velocity is v, its final velocity is $0\,\mathrm{m/s}$, and the braking distance is s. The maximum braking force is provided by the static friction force between the tires and the road. On a horizontal road, the normal force is $N = mg$, therefore the maximum static friction force is

$$F_s = \mu N = \mu mg.$$

According to Newton's second law, this force causes the deceleration of the vehicle:

$$ma = \mu mg.$$

After cancelling the mass m, we obtain

$$a = \mu g.$$

Rigid Bodies XXII

The uniformly decelerated motion can also be described by the kinematic relation

$$v_f^2 = v_i^2 - 2as.$$

Here the final velocity is $v_f = 0\,\mathrm{m/s}$ and the initial velocity is $v_i = v$, therefore

$$0 = v^2 - 2as.$$

Solving this equation for the magnitude of the deceleration gives

$$a = \frac{v^2}{2s}.$$

Since the same deceleration is produced by the maximum static friction force, we can equate the two expressions for a:

$$\mu g = \frac{v^2}{2s}.$$

Rigid Bodies XXIII

Therefore, the coefficient of static friction is

$$\mu = \frac{v^2}{2gs}.$$

Now consider the same fully braked vehicle on an inclined road with slope angle α. The component of the gravitational force parallel to the slope is

$$mg \sin \alpha.$$

The normal force perpendicular to the slope is

$$N = mg \cos \alpha.$$

The maximum static friction force is therefore

$$F_s = \mu N = \mu mg \cos \alpha.$$

Rigid Bodies XXIV

For the vehicle to remain at rest on the slope, the downslope component of gravity must be balanced by the static friction force:

$$mg \sin \alpha = \mu mg \cos \alpha.$$

After cancelling mg, we get

$$\sin \alpha = \mu \cos \alpha.$$

Dividing both sides by $\cos \alpha$ gives

$$\tan \alpha = \mu.$$

Therefore, the slope angle is

$$\alpha = \arctan \mu.$$

Rigid Bodies XXV

Substituting the expression for μ obtained from the braking distance:

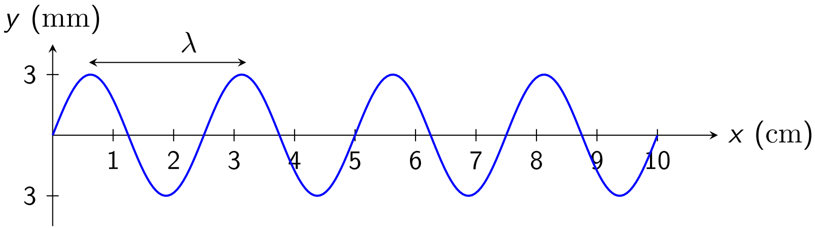
$$\alpha = \arctan\left(\frac{v^2}{2gs}\right).$$

Hence, the required slope angle is

$$\boxed{\alpha = \arctan\left(\frac{v^2}{2gs}\right)}.$$

Wave Optics I

1. The wave shown in the figure was generated by a 60 Hz resonator. Determine:
   a) amplitude of the wave
   b) frequency of the wave
   c) wavelength of the wave
   d) propagation speed of the wave
   e) period of the wave



Wave Optics II

(a) From the graph, the maximum displacement is

$$A = 3\,\mathrm{mm} = 3 \times 10^{-3}\,\mathrm{m}.$$

Therefore,

$$\boxed{A = 3\,\mathrm{mm}}.$$

(b) The wave is generated by a resonator operating at

$$f = 60\,\mathrm{Hz} = 60\,\mathrm{s}^{-1}.$$

Hence,

$$\boxed{f = 60\,\mathrm{Hz}}.$$

Wave Optics III

(c) The wavelength is the spatial length of one complete cycle of the wave:

$$\lambda = 2\,\mathrm{cm} = 0.02\,\mathrm{m}.$$

Hence,

$$\boxed{\lambda = 2\,\mathrm{cm}}.$$

(d) The propagation speed of a wave is

$$v = \lambda f.$$

Substitute the numerical values:

$$v = (0.02\,\mathrm{m})(60\,\mathrm{s}^{-1}) = 1.2\,\mathrm{m/s}.$$

Wave Optics IV

Therefore,

$$\boxed{v = 1.2\,\mathrm{m/s}}.$$

(e) The period is the reciprocal of the frequency:

$$T = \frac{1}{f} = \frac{1}{60\,\mathrm{s}^{-1}} = 0.0167\,\mathrm{s}.$$

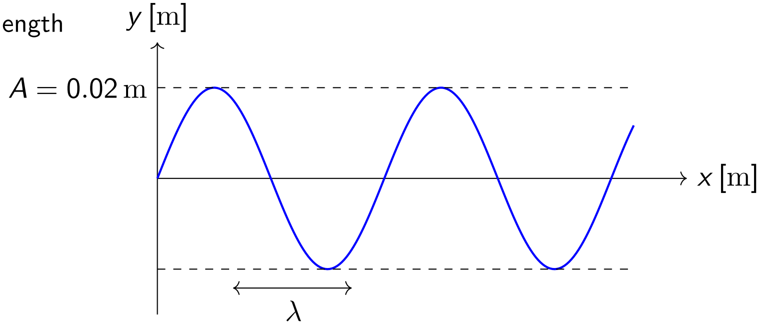
Therefore,

$$\boxed{T = 0.0167\,\mathrm{s}}.$$

Wave Optics V

2. A wave propagating along a string can be defined as follows (with x in meters, t in seconds): $y(x, t) = 0.02\,\mathrm{m}\sin(30\,\mathrm{s}^{-1}t - 4\,\mathrm{m}^{-1}x)$. Determine the:

   a) amplitude
   b) frequency
   c) speed
   d) wavelength



Wave Optics VI

Compare the given equation

$$y(x,t) = 0.02\sin(30t - 4x)$$

with the standard form of a harmonic traveling wave:

$$y(x,t) = A\sin(\omega t - kx).$$

From this comparison, we identify

$$A = 0.02\,\mathrm{m}, \qquad \omega = 30\,\mathrm{rad/s}, \qquad k = 4\,\mathrm{rad/m}.$$

(a) The amplitude is directly obtained from the equation:

$$\boxed{A = 0.02\,\mathrm{m}}.$$

Wave Optics VII

(b) The angular frequency and the frequency are related by

$$\omega = 2\pi f.$$

Therefore,

$$f = \frac{\omega}{2\pi} = \frac{30}{2\pi} = 4.77\,\mathrm{Hz}.$$

Hence,

$$\boxed{f = 4.77\,\mathrm{Hz}}.$$

Wave Optics VIII

(c) The wave speed is

$$v = \frac{\omega}{k}.$$

Substituting the numerical values:

$$v = \frac{30\,\mathrm{rad/s}}{4\,\mathrm{rad/m}} = 7.5\,\mathrm{m/s}.$$

Therefore,

$$\boxed{v = 7.5\,\mathrm{m/s}}.$$

Wave Optics IX

(d) The wavelength is related to the wave number by

$$k = \frac{2\pi}{\lambda}.$$

Therefore,

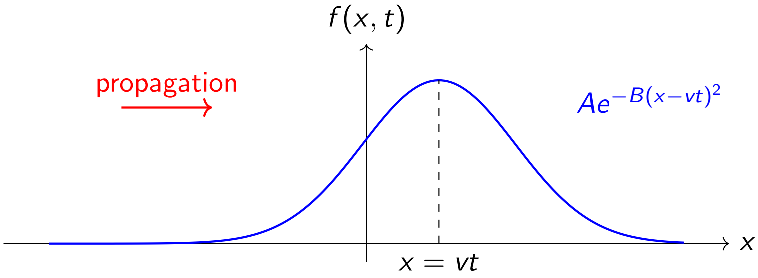
$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{4} = 1.57\,\mathrm{m}.$$

Hence,

$$\boxed{\lambda = 1.57\,\mathrm{m}}.$$

Wave Optics X

3. Consider a wave given by the equation $f(x,t) = Ae^{-B(x-vt)^2}$. Prove that this wave equation satisfies the general one-dimensional wave equation.



Wave Optics XI

Let $u = x - vt$. Then the wave can be written as

$$f(x,t) = Ae^{-Bu^2}.$$

First calculate the spatial derivative:

$$\frac{\partial f}{\partial x} = -2AB(x - vt)e^{-B(x-vt)^2}.$$

The second spatial derivative is

$$\frac{\partial^2 f}{\partial x^2} = \left[-2AB + 4AB^2(x - vt)^2\right] e^{-B(x-vt)^2}.$$

Now calculate the time derivative:

$$\frac{\partial f}{\partial t} = 2ABv(x - vt)e^{-B(x-vt)^2}.$$

Wave Optics XII

The second time derivative is

$$\frac{\partial^2 f}{\partial t^2} = v^2 \left[-2AB + 4AB^2(x - vt)^2\right] e^{-B(x-vt)^2}.$$

Comparing the two second derivatives gives

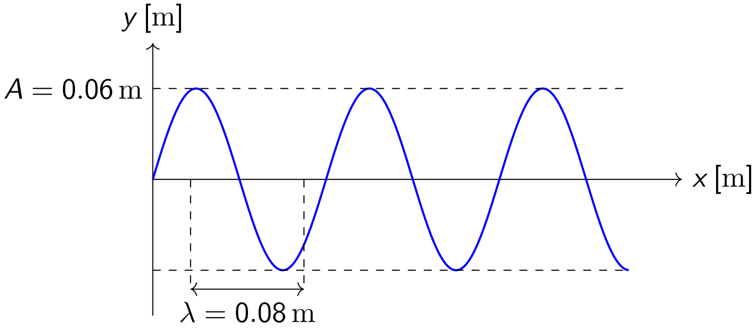
$$\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial x^2}.$$

Therefore, the function satisfies the one-dimensional wave equation

$$\boxed{\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial x^2}}.$$

Wave Optics XIII

4. Determine the amplitude, frequency, and wavelength of the wave shown in the figure below, if the propagation speed of the wave is 300 m/s. Write a solution for the wave if at t=0 s the wave appears as shown in the figure.



Wave Optics XIV

From the graph,

$$\boxed{A = 0.06\,\mathrm{m}}.$$

From the graph, the length corresponding to $\frac{5}{2}\lambda$ is $20\,\mathrm{cm}$:

$$\frac{5}{2}\lambda = 20\,\mathrm{cm}.$$

Therefore,

$$\lambda = \frac{2 \cdot 20\,\mathrm{cm}}{5} = 8\,\mathrm{cm} = \boxed{0.08\,\mathrm{m}}.$$

The relation between wave speed, wavelength, and frequency is

$$v = f\lambda.$$

Wave Optics XV

Therefore,

$$f = \frac{v}{\lambda} = \frac{300\,\mathrm{m/s}}{0.08\,\mathrm{m}} = 3750\,\mathrm{Hz}.$$

Thus,

$$\boxed{f = 3750\,\mathrm{Hz}}.$$

The general form of a harmonic traveling wave is

$$y(x,t) = A\sin(\omega t - kx).$$

The angular frequency is

$$\omega = 2\pi f = 2\pi \cdot 3750\,\mathrm{Hz} \approx 23600\,\mathrm{rad/s}.$$

Wave Optics XVI

The wave number is

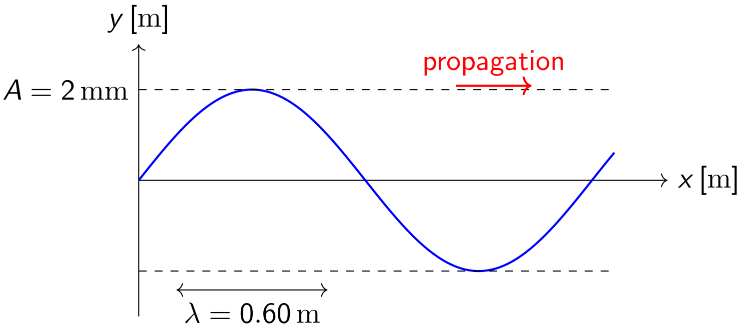
$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.08} \approx 78.5\,\mathrm{rad/m}.$$

Substituting the parameters gives

$$\boxed{y(x,t) = 0.06\,\mathrm{m}\ \sin(23600\,\mathrm{rad\,s^{-1}}\ t - 78.5\,\mathrm{rad\,m^{-1}}\ x)}.$$

Wave Optics XVII

5. For a wave propagating along a string with a frequency of 30 Hz, $\lambda = 60\,\mathrm{cm}$, A=2 mm, write the solution in SI units.



Wave Optics XVIII

First convert all quantities into SI units:

$$\begin{aligned} f = 30\,\mathrm{Hz} &= 30\,\mathrm{s}^{-1}, \\ \lambda = 60\,\mathrm{cm} &= 0.60\,\mathrm{m}, \\ A = 2\,\mathrm{mm} &= 2 \cdot 10^{-3}\,\mathrm{m}. \end{aligned}$$

The general form of a harmonic wave propagating in the positive x direction is

$$y(x,t) = A \sin(\omega t - kx).$$

The angular frequency is

$$\omega = 2\pi f = 2\pi \cdot 30 = 60\pi\,\mathrm{rad/s}.$$

Wave Optics XIX

The wave number is

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.60} = 10.47\,\mathrm{rad/m}.$$

Substituting the values into the wave equation gives

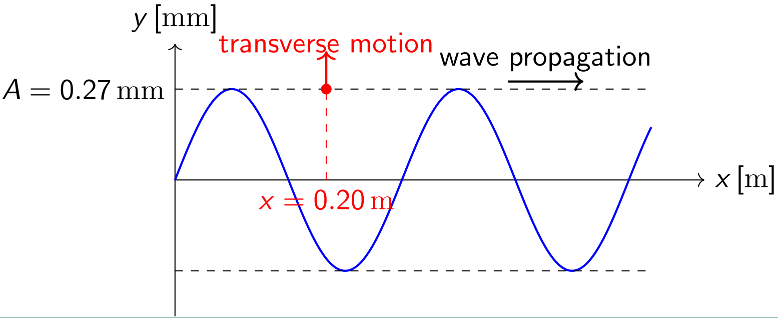
$$y(x,t) = 2 \cdot 10^{-3}\,\mathrm{m}\,\sin(60\pi\,\mathrm{s}^{-1}\,t - 10.47\,\mathrm{m}^{-1}\,x).$$

Therefore, the wave function in SI units is

$$\boxed{y(x,t) = 2 \cdot 10^{-3}\sin(60\pi\,\mathrm{s}^{-1}\,t - 10.47\,\mathrm{m}^{-1}\,x)\,\mathrm{m}}.$$

Wave Optics XX

6. A wave solution is given by $y = 0.27\sin(12x - 500t)\ mm$, where x is in meters and t is in seconds. For the transverse wave at x=20 cm, determine:
   a) its speed
   b) its acceleration
   c) What is the displacement of the point at t=4 s?



Wave Optics XXI

First convert the amplitude to SI units:

$$A = 0.27\,\mathrm{mm} = 0.27 \cdot 10^{-3}\,\mathrm{m} = 2.7 \cdot 10^{-4}\,\mathrm{m}.$$

Thus the wave can be written as

$$y(x, t) = 2.7 \cdot 10^{-4} \sin(12\,\mathrm{s}^{-1}\,x - 500\,\mathrm{m}^{-1}\,t)\,\mathrm{m}.$$

At the selected point,

$$x = 0.20\,\mathrm{m},$$

the phase becomes

$$12\,\mathrm{m}^{-1}\,x - 500\,\mathrm{s}^{-1}\,t = 12 \cdot 0.20 - 500\,\mathrm{s}^{-1}\,t = 2.4 - 500\,\mathrm{s}^{-1}\,t.$$

Wave Optics XXII

Therefore,

$$y(0.20\,\mathrm{m}, t) = 2.7 \cdot 10^{-4} \sin(2.4 - 500\,\mathrm{s}^{-1}\, t)\,\mathrm{m}.$$

(a) **Transverse velocity:** The transverse velocity is the time derivative of the displacement:

$$v_y(t) = \frac{\partial y}{\partial t}.$$

Differentiate:

$$v_y(t) = 2.7 \cdot 10^{-4} \cdot (-500\,\mathrm{s}^{-1})\,\mathrm{m}\,\cos(12\,\mathrm{m}^{-1}\, x - 500\,\mathrm{s}^{-1}\, t).$$

Hence,

$$v_y(x, t) = -0.135 \cos(12\,\mathrm{m}^{-1}\, x - 500\,\mathrm{s}^{-1}\, t)\,\mathrm{m/s}.$$

Wave Optics XXIII

At $x = 0.20\,\mathrm{m}$:

$$\boxed{v_y(t) = -0.135\cos(2.4 - 500\,\mathrm{s}^{-1}\ t)\,\mathrm{m/s}}.$$

(b) **Acceleration:** The acceleration is the time derivative of the transverse velocity:

$$a_y(t) = \frac{\partial v_y}{\partial t}.$$

Differentiate:

$$\boxed{a_y(t) = -67.5\sin(2.4 - 500,\mathrm{s}^{-1}\ t)\,\mathrm{m/s^2}}.$$

Wave Optics XXIV

(c) **Displacement at $\mathbf{t = 4\,s}$:** Substitute $t = 4\,\mathrm{s}$ into the displacement function:

$$y(0.20\,\mathrm{m}, 4\,\mathrm{s}) = 0.27\sin(12\,\mathrm{m}^{-1} \cdot 0.20\,\mathrm{m} - 500\,\mathrm{s}^{-1} \cdot 4\,\mathrm{s})\,\mathrm{mm}.$$

This gives

$$y(0.20\,\mathrm{m}, 4\,\mathrm{s}) = 0.27\sin(2.4 - 2000)\,\mathrm{mm} \approx \boxed{0.118\,\mathrm{mm}}.$$

The End

Thank you for your attention!